

Development and Control of Articulated Mobile Robot for Climbing Steep Stairs

Motoyasu Tanaka^{ID}, *Member, IEEE*, Mizuki Nakajima, Yosuke Suzuki^{ID}, *Member, IEEE*, and Kazuo Tanaka^{ID}, *Fellow, IEEE*

Abstract—In this paper, we develop an articulated mobile robot that can climb stairs, and also move in narrow spaces and on 3-D terrain. This paper presents two control methods for this robot. The first is a 3-D steering method that is used to adapt the robot to the surrounding terrain. In this method, the robot relaxes its joints, allowing it to adapt to the terrain using its own weight, and then, resumes its motion employing the follow-the-leader method. The second control method is the semi-autonomous stair climbing method. In this method, the robot connects with the treads of the stairs using a body called a connecting part, and then shifts the connecting part from its head to its tail. The robot then uses the sensor information to shift the connecting part with appropriate timing. The robot can climb stairs using this method even if the stairs are steep, and the sizes of the riser and the tread of the stairs are unknown. Experiments are performed to demonstrate the effectiveness of the proposed methods and the developed robot.

Index Terms—Articulated mobile robot, plant inspection, search and rescue, stair climbing, terrain adaptability.

I. INTRODUCTION

ARTICULATED mobile robots have structures in which multiple segments that can generate a propulsive force are connected by joints, which allows the robots to have elongated shapes and to enter narrow spaces. In addition, such robots can operate on the complex terrain because they move under the effect of propulsive forces that are generated by the segments while freely changing their body shapes via their joints. Many articulated mobile robots have been developed (see [1]) for applications such as pipe inspection [2]–[5] and search and rescue operations at disaster sites [6]–[9] because of these characteristics.

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M. Tanaka, M. Nakajima, and K. Tanaka are with the Department of Mechanical Intelligent Systems Engineering, University of Electro-Communications, Tokyo 182-8585, Japan (e-mail: mtanaka@uec.ac.jp; mizuki.nakajima@rc.mce.uec.ac.jp; ktanaka@mce.uec.ac.jp).

Y. Suzuki is with Kanazawa University, Kanazawa 920-1192, Japan (e-mail: suzuki@se.kanazawa-u.ac.jp).

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Considering the funds, operation, and maintenance, it is important how to use a rescue robot in nondisaster situations. One of the use of a rescue robot other than disaster scenarios is inspection of industrial plants. Mobility required for plant inspection robots is ascending and descending of steep stairs, and overpassing obstacles such as piping. The crawler type with flippers and leg type robots have been developed for gas and oil plants as [10]. It seems that an articulated mobile robot can make an inspection through a narrow passage that could not be passed by the other type robots because it has elongated bodies. The ultimate goal of this study is to realize an articulated mobile robot that is capable of not only search and rescue operations at disaster sites but also inspection of industrial plants in nondisaster situations.

Stairs are an example of the terrain on which it is difficult for the articulated mobile robots to move. A problem that typically arises when an articulated mobile robot climbs stairs is that a stuck state occurs for the joints and the edges of the stairs because most articulated mobile robots do not have propulsion mechanisms located in their joints. The ACM-R4.1 [11] and 4.2 [12], which are robots composed of rotary joints and active wheels that are equipped with torque sensors, can enter narrow areas because they are relatively small, but no stair climbing ability has been reported for these robots. In contrast, the KR-II [13] was able to climb stairs using large-diameter active wheels and joints that move up and down linearly. The ACM-R8 [14], which is composed of joints with a parallel link mechanism and large-diameter monotread wheels, was also able to climb stairs. KR-II and ACM-R8 were able to climb stairs by arranging their joints so that they did not come into contact with the terrain and used the capabilities of large-diameter wheels to overcome high obstacles.

One way to avoid the stuck state that results from contact between the joints and stairs is to reduce the proportion of the robot's joints along the body length direction while increasing the proportion of the segments that provide propulsive forces. All parts of the flexible monotread mobile track [15] can generate a propulsive force along the body length direction, and the track can propel the robot as a whole, including the bending part; the robot, therefore, can succeed in climbing both linear stairs and a spiral staircase. Another way to avoid the stuck state is the appropriate use of joint motion. The KAIRO 3 [8] robot has modules with relatively small active wheels that are connected by a three-degree-of-freedom joint that includes an oblique swivel axis. The KAIRO 3 has a 3-D laser sensor on

its head, and autonomously climbed two steps on a staircase using a map created from 3-D scans. However, the staircase in this example consisted of only two steps. The OT-4[9] robot has segments with tracks that are serially connected using two-degree-of-freedom rotary joints. The OT-4 climbed stairs that sloped at 40° when supported by a reinforcement learning system called the 7G control system [16]. The joint motion required to avoid a stuck formation is obtained by the learning system.

In the cited studies, each robot climbed stairs while pressing a wheel or crawler against the edge of a staircase. Therefore, when the stairs are steep, slippage occurs and it is impossible for the robot to climb these stairs.

This paper aims to realize an articulated mobile robot that is capable of not only movement in narrow spaces and 3-D movement for search and rescue at disaster sites but also climbing steep stairs for inspecting a plant. This paper deals with two control problems of an articulated mobile robot. One is terrain-following control on the uneven terrain for search and rescue tasks. The other is stair climbing control for inspection tasks in nondisaster situation.

The contributions of this paper are as follows.

- 1) A 3-D steering method that uses manual intermittent terrain adaptation is proposed. The robot adapts to the surrounding terrain by zeroing the torque of some of its joints, and it then moves while following the terrain by resuming 3-D motion from the posture that is suited to the terrain. The method does not require either the measurement of forces (e.g., the wheel torque and external force) or the measurement of external information (e.g., distances to obstacles). Because torque and force measurements are unnecessary, even a simple robot that is composed of a servo motor for hobby purposes can adapt to the surrounding terrain using the proposed method, and the cost of the robot can thus be reduced.
- 2) A semi-autonomous stair climbing control method that uses sensor information is proposed. A simple operation by the operator allows the robot to ascend/descend staircases with unknown height and depth parameters. The effectiveness of the proposed method does not depend on the presence or absence of a riser. The robot can climb the stairs safely, even if the stairs are steep.
- 3) An articulated mobile robot for stair climbing is developed, the basic performance of the robot is evaluated, and the effectiveness of the proposed control method is demonstrated through use of the robot. While the robot is relatively small, it has high mobility and is able to climb a steep step sloped at 54.5° and a 1-m-high step.

The remainder of this paper is organized as follows. Section II describes the robot model used and the basic 3-D steering control method and a simple terrain adaptation method that is based on the follow-the-leader (FTL) approach. Section III proposes a control method that enables the robot to climb up and down stairs semi-autonomously. Section IV describes an articulated mobile robot that was developed to realize 3-D motion and stair climbing. Section V evaluates the performance of the developed robot and demonstrates the effectiveness of the proposed control method experimentally using this robot. Section VI concludes this paper.

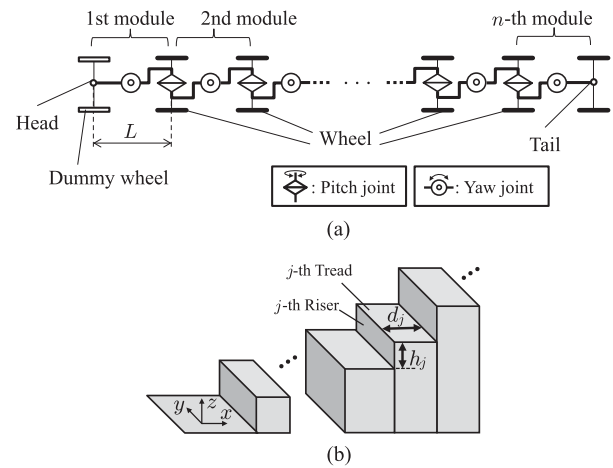


Fig. 1. Models of the robot and the stairs. (a) Articulated mobile robot. (b) Stairs.

II. MODEL AND 3-D STEERING CONTROL

It is difficult to control articulated mobile robots because they have many degrees of freedom. Two main methods are used to control these robots [1]: n-trailer and FTL. n-trailer, which is used in [6] and [17], is a method by which the robot's motion is calculated in order from head to tail while assuming that any segment that generates propulsion does not skid. This method is only applied to planar terrain. There is also a similar method by which the motions of all joints and wheels are calculated simultaneously using a kinematic model that is derived by assuming that any wheel mounted on the segment does not skid [18]–[22]. This method was mainly proposed for use with snake robots in which links with passive wheels are serially connected through active joints. When using this method, it is possible to consider the case that some of the robot's wheels are lifted and do not touch the ground, and it is also possible to track the trajectory of the head position and the orientation simultaneously while avoiding obstacles by switching the grounding status of the wheels, as in [18] and [19]. However, only steps [20], [21] and cylindrical surfaces [22] are treated as 3-D terrains in this case, and a control method for more complex 3-D terrain has not yet been established. In contrast, the FTL method used in [2], [3], and [11]–[14] allows the head (leader) movement to be shifted from head to tail. Using this method, the operator can easily perform 3-D operations by appropriate design of the robot's head movement. In this section, we propose a simple terrain adaptation method that is based on the FTL approach.

A. Model

Fig. 1(a) shows the model of the articulated mobile robot. The joint uses a configuration in which the yaw rotational joint and the pitch rotational joint are connected alternately in series and wheels are mounted on the left and right sides coaxially with respect to the pitch joint. This is the same joint configuration that was used in the ACM-R4.2 [12], and allows various types of motion control; e.g., control for moving-obstacle avoidance [18], whole-body collision avoidance [19], step climbing [20], [21], and stair climbing [23]. The robot can be propelled by

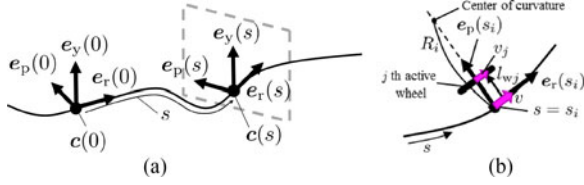


Fig. 2. Definition of a continuous backbone curve. (a) Coordinates. (b) Velocity of a wheel.

rotating the wheels while simultaneously varying the body shape using the pitch and yaw joints. Dummy wheels with low friction and easy side slip are attached at the head of the robot. The purpose of these wheels is to simplify stair climbing control by maintaining a constant distance between the head and tread surface when the robot climbs a stair. Let l be the link length, $L = 2l$ be the module length, r be the wheel radius, ϕ_i be the angle of the i th yaw joint, and ψ_i be the angle of the i th pitch joint, and assume that the number of modules n is sufficiently large.

B. Shape Fitting Using the Backbone Curve

We use the shape fitting method that was proposed by Yamada *et al.* [11], [33] for basic 3-D steering of the robot because the curve representation in their method is intuitive and suitable for a robot that has the joint configuration shown in Fig. 1(a); additionally, the method has low computational cost for calculation of the target angles of the joints. A differential equation is introduced as a continuous backbone curve to express the target robot body shape

$$\begin{cases} \frac{dc(s)}{ds} = e_r(s) \\ \frac{de_r(s)}{ds} = \kappa_y(s)e_p(s) - \kappa_p(s)e_y(s) \\ \frac{de_p(s)}{ds} = -\kappa_y(s)e_r(s) \\ \frac{de_y(s)}{ds} = \kappa_p(s)e_r(s) \end{cases} \quad (1)$$

where $c = [x(s), y(s), z(s)]^T$ is a vector that represents the coordinates of the continuous curve, and s is the length variable along the curve as shown in Fig. 2(a). $(e_r(s), e_p(s), e_y(s))$ is the backbone curve reference set that is used as the orthogonal coordinate system, $e_r(s)$ is a tangential unit vector for the curve at s , and $e_p(s)$ and $e_y(s)$ are unit vectors that are oriented along the pitch axis and the yaw axis, respectively. $\kappa_p(s)$ and $\kappa_y(s)$ are the curvatures around the pitch axis and the yaw axis in the curve reference set, respectively. $\kappa_p(s)$ and $\kappa_y(s)$ are parameters that are determined by the operator. $s = s_h$ is the position of the head of the robot on the curve, and the tail is located at $s = s_h - nL$.

The robot joint angles ϕ_i and ψ_i can be obtained approximately using κ_y and κ_p as

$$\begin{aligned} \phi_i &= - \int_{s_h - 2il}^{s_h - 2(i-1)l} \kappa_y(s) ds \\ \psi_i &= - \int_{s_h - (2i+1)l}^{s_h - 2(i-1)l} \kappa_p(s) ds. \end{aligned} \quad (2)$$

Note that the sign of (2) is different from that in [11] and [33] because the beginning and end positions of the curves are different.

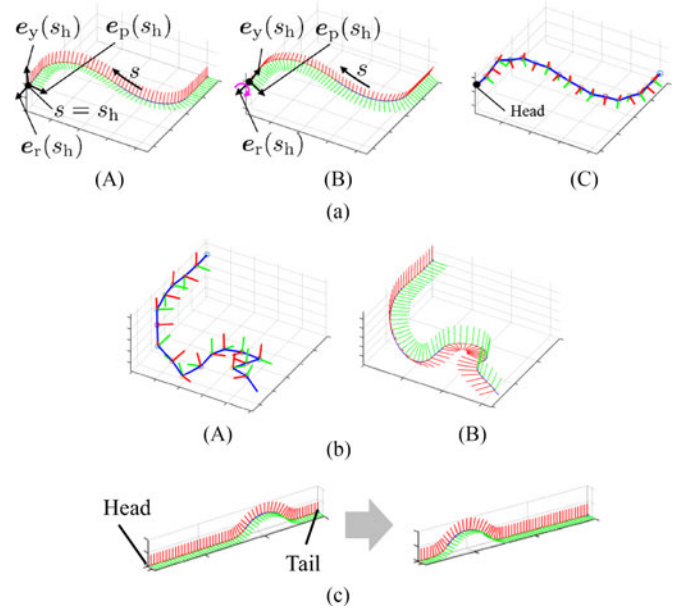


Fig. 3. Continuous target curve and robot posture. (a) Shape fitting method. (A) Target continuous backbone curve; (B) target continuous backbone curve after lateral rolling; (C) robot posture calculated using (2). In (C), red lines depict the rotational axis vectors for each joint. (b) Robot posture and continuous curve used in terrain adaptation. (A) Posture of the robot; (B) target continuous curve calculated using (5). (c) Continuous curve used in recovery motion from a stuck state.

By varying s_h , the area of the robot body on the curve changes, and FTL-type control is performed along the continuous curve.

Additionally, the robot can perform a lateral rolling motion by rotating $\kappa_p(s)$ and $\kappa_y(s)$ uniformly around e_r for the entire curve. Let Ψ be the rotational angle around e_r . $\kappa_p(s)$ and $\kappa_y(s)$ can then be calculated as

$$\begin{bmatrix} \kappa_y(s) \\ \kappa_p(s) \end{bmatrix} = \begin{bmatrix} \cos \Psi & -\sin \Psi \\ \sin \Psi & \cos \Psi \end{bmatrix} \begin{bmatrix} \kappa'_y(s) \\ \kappa'_p(s) \end{bmatrix} \quad (3)$$

where $\kappa'_p(s)$ and $\kappa'_y(s)$ represent $\kappa_p(s)$ and $\kappa_y(s)$ before the rolling motion, respectively. The robot can perform lateral rolling motion when the operator changes Ψ as in Fig. 3(a).

C. Design of Wheel Velocities

The wheel is located at a position that has deviated from the target continuous curve. Let us define the wheel velocity as $ds_h/dt = v$. It is then necessary to correct the velocity of the active wheel by considering the offset caused by the curve and the curvature of the continuous curve. Let v_j be the translational velocity of the j th active wheel when mounted coaxially with the i th pitch joint, $s = s_i$ be the position of the i th pitch joint located on the target continuous curve, and l_{wj} be the offset length between $c(s_i)$ and the center of the j th active wheel. For the continuous curve, e_p and the axle of the wheels are located on the same straight line. As shown in Fig. 2(b), if $\kappa_y(s_i) \neq 0$, then the distance to the center of the curvature at $s = s_i$ is $R_i = \frac{1}{\kappa_y(s_i)}$ on the plane that is defined by $e_r(s_i)$ and $e_p(s_i)$. Therefore, according to the geometric relation shown in

Fig. 2(b), v_j can be obtained as

$$v_j = \frac{(R_i - l_{wj})}{R_i} v. \quad (4)$$

D. Steering Method

The operator first varies the continuous curve at the head of the robot by manipulating κ_y and κ_p in the region of the head. FTL-type control is then performed by varying the position of the robot along the continuous curve; i.e., by changing v , which is the velocity of s_h . In addition, it is possible to perform a lateral rolling motion by varying Ψ . In this case, the operator changes Ψ , κ_p , and κ_y are calculated using (3), and the target joint angles for the lateral rolling motion are calculated using (2). The active wheel rotation speed is then calculated using (4). Therefore, 3-D steering of the robot can be realized by controlling κ_p , κ_y , v , and Ψ .

E. Terrain-Following by Manual Intermittent Adaptation

To move across complex terrain, a robot requires operating modes such as terrain following, obstacle avoidance, and utilization (pushing) of obstacles. Considerable research into these operations has been carried out on both articulated mobile robots and snake-like robots without wheels. On a 2-D plane, adaptation to obstacles based on the contact force with these obstacles [24], obstacle-aided locomotion [25]–[27], and obstacle avoidance based on distance sensor information [19], [28] have all been proposed. Motion inside and outside a pipe, adaptation to changes in pipe diameter [29] and bent pipes [30] through pipe diameter estimation using an extended Kalman filter, and motion on a horizontal pipe using a passive compliant element to adapt to flange and pipe diameter changes [31] have been proposed. On complex 3-D terrain, adaptation to the terrain using a torque sensor-based method [11], [32] has been proposed. If all the following parts of the robot can continue along the path of the head of the robot when using FTL-type control, the robot can move forward, even across the complex 3-D terrain. However, in practice, wheel slip and changes in the contact position between the robot and terrain cause the following parts to travel along a path that differs from that of the head. This causes problems, including unintentional contact with the terrain resulting in a stuck state and excessive loading of the joints, and may even cause the robot to fall. By adapting to the terrain in the manner of [11], [32], the robot can reduce joint loading and prevent itself from falling.

This paper proposes a simple terrain adaptation method that does not require either measurement of forces, e.g., the wheel torque and the external force, or measurement of external information, e.g., distances to obstacles. Because torque and force measurements are unnecessary, even a simple robot that is composed of a servo motor for hobby purposes can adapt to the surrounding terrain using the proposed method, and the cost of the robot can thus be reduced. In this method, the robot makes the torque at the joint zero, and adjusts its shape to that of the terrain using its own weight. The problem to be solved here is how to calculate the continuous curve c after the terrain adaptation process. FTL operation is performed using the joint angle, which

is calculated from the continuous curve c . When the torque at the joint is set to zero, the joint angle changes because of the interaction with the terrain, and the entire continuous curve must be recalculated from the perspective of the current angle. Because determination of a continuous curve from discrete angle information is an inverse problem, the solution is not determined uniquely. We, therefore, assume that the curvature $\kappa_y(s)$ at $s_h - 2il < s \leq s_h - 2(i-1)l$ and the curvature $\kappa_p(s)$ at $s_h - (2i+1)l < s \leq s_h - (2i-1)l$ are the constant values κ_{yi} and κ_{pi} , respectively. From (2), κ_{yi} and κ_{pi} can then be calculated as

$$\kappa_{yi} = -\frac{\phi_i}{2l}, \quad \kappa_{pi} = -\frac{\psi_i}{2l}. \quad (5)$$

The curvatures $\kappa_y(s)$ and $\kappa_p(s)$ can be obtained by calculating (5) for all joints, and the robot can then restart FTL operation based on the adapted body shape after terrain adaptation. An example is shown in Fig. 3(b). It is thus considered that the robot can respond flexibly to complex terrain by simply varying the timing and the corresponding joints when the operator sets the torque to zero, depending on the specific situation. Note that there are fall risks that depend on the choice of joint when the torque is set to zero if there are few contact points between the robot and terrain (e.g., when overcoming a valley).

F. Recovery Motion From a Stuck State

If the terrain is not smooth, the joint parts of the robot may become stuck even if the robot uses the terrain-following method. Thus, a traveling wave motion of the pitch angles is used as a recovery motion from a stuck state. In the motion, a traveling wave moving from the tail to the head is prepared on κ_p as in Fig. 3(c), and the robot moves using a curve obtained by overlaying a traveling wave on the original continuous backbone curve. If the traveling wave reaches the stuck part of the body, the stuck part separates from the terrain and the robot recovers from the stuck state.

III. SEMI-AUTONOMOUS STAIR CLIMBING

The stairs shown in Fig. 1(b) are assumed to be the environment. Let h_j be the height of the j th riser and let d_j be the depth of the j th tread. It is assumed that the robot can detect contact between each pair of wheels and the stair tread surface using sensors that are mounted on the robot.

A sufficiently large value of d_j with respect to L makes the motion planning process easier because the area in which the robot can move is then wider. In contrast, if d_j is small, then the area for the robot is small. In this case, few wheels are touching the tread and the risk of the robot falling increases when some of these wheels fall from the tread. In addition, the body of the robot cannot move far from the riser because d_j is small. It is thus difficult for the robot to climb the stairs while avoiding collisions between the robot and stairs. We previously proposed a stair climbing method for the case that d_j is relatively small in [23]. However, this method assumes that the size of the rise/tread of the stairs is known, and the operator must, therefore, check the relative relationship between the robot and stairs from a

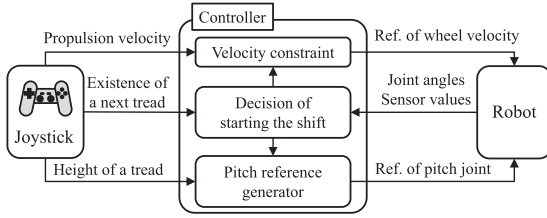


Fig. 4. Control diagram of the semi-autonomous stair climbing process.

bird's-eye viewpoint and give commands at appropriate times. This situation is unrealistic and impractical.

This paper proposes a control method for an articulated mobile robot on stairs with unknown riser and tread parameters that improves on the previous method [23]. Assuming that remote control is used, a camera is normally installed at the head of the robot. Even when the height of the rise is unknown, it is possible to adjust the height to an extent by operating the robot on the basis of the camera images and sensor information. It is therefore assumed that the operator sets the rise height manually using the sensor information. In contrast, if the depth of the tread is unknown, it then becomes difficult for the operator to operate the robot. This is because it is difficult for the operator to recognize the tread depth and to measure the depth using a sensor (e.g., a distance sensor) when there is no riser surface present. In the proposed method, the operator does not need to consider the tread depth and can operate the robot by focusing on the head alone, because the robot moves semi-autonomously according to sensor information from the body part.

Fig. 4 shows the control diagram for the proposed method. The operator provides the robot with the following three control commands using a joystick:

- 1) the propulsion velocity;
- 2) the decision on whether or not the next stage exists;
- 3) adjustment of the estimated height of the next riser that the head is approaching;

In the pitch reference generator, the target pitch joint angle is calculated employing the method described in Section III-A and is then output to the robot. In the start condition for the shift, the decision to start the shift that is related to the target pitch angle is made according to the conditions that are described in Section III-B. In velocity constraint, a velocity constraint is determined following Section III-D, and the target wheel speed is determined. The case of climbing up stairs is described first in this section, while the case of climbing down stairs is described in Section III-E.

A. Shift in the Connecting Part [23]

In [23], the connecting part, which is the module of the robot that connects the two treads, was introduced and the robot climbed up stairs by shifting the connecting part from its head to its tail. If the robot uses the shift in the connecting part, then the yaw angle of the connecting part must be zero, as per [20] and [23]. The yaw angle thus cannot be used to steer the robot. In the case that d_j is small, almost all modules become a connecting part having a zero yaw angle and the robot thus cannot use the

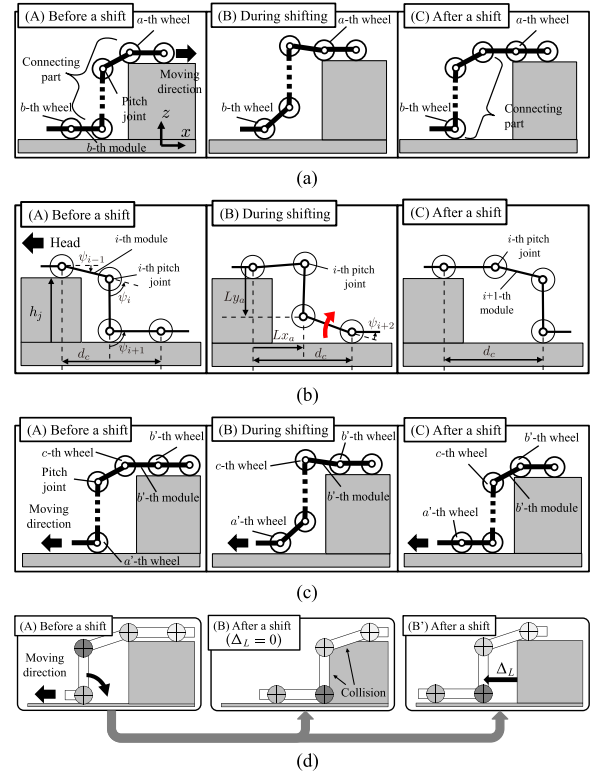


Fig. 5. Shift in the connecting part. (a) Climbing up the stairs. (b) Climbing up the stairs (two modules). (c) Climbing down the stairs. (d) Shift with Δ_L in climbing down stairs.

yaw angles for steering purposes. We proposed a straight stair climbing method without steering in [23] while assuming that d_j was small. The state of the shift in the connecting part from [23] is shown in Fig. 5(a). By maintaining a constant relative distance between the a th wheel and the b th wheel, the robot can prevent these wheels from falling out of the tread during the shift because the following part of the connecting part will not travel backwards. When using this method, the robot cannot steer when climbing up and down the stairs, but it can at least climb the narrow stairs.

Here, we consider the case that the connecting part consists of two modules. Fig. 5(b) depicts the shift in the connecting part in the case of climbing up the stairs. The i th and $(i + 1)$ th modules represent the connecting part before the shift. After the shift, the connecting part is represented by the $(i + 1)$ th and $(i + 2)$ th modules. The connecting part connects the j th and $(j - 1)$ th treads, and we also define $L \leq h_j \leq 2L$ here. Let the start time of the shift be t_{st} and the end time of the shift be $t_{st} + t_{\psi}$. We set x_a and y_a as

$$x_a = (d_c - L \cos \psi_{i+2})/L, \quad y_a = (L \sin \psi_{i+2} - h_j)/L. \quad (6)$$

From Fig. 5(b), the geometric relationships are

$$d_c = L + \sqrt{L^2 - (h_j - L)^2} \quad (7)$$

$$Lx_a = -L\{\sin \psi_{i-1} + \sin(\psi_{i-1} + \psi_i)\} \quad (8)$$

$$Ly_a = L\{\cos \psi_{i-1} + \cos(\psi_{i-1} + \psi_i)\}. \quad (9)$$

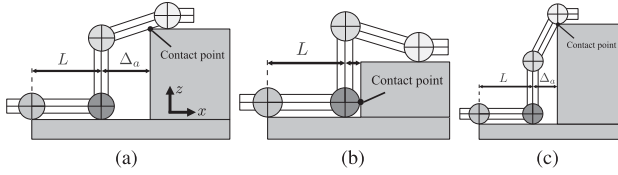


Fig. 6. Δ_a and the connecting part (two modules) for (a) $\Delta_a > r$, (b) $\Delta_a = r$, and (c) highest riser. If the riser is higher than that shown in (c), then wheel a cannot be in contact with the tread because of contact between the body of the robot and the tread.

From (7)–(9), we set the motions of the pitch joints $\psi_{i-1}, \dots, \psi_{i+2}$ as

$$\left. \begin{aligned} \psi_{i+2} &= \frac{\pi}{2} t' \\ \psi_i &= -\cos^{-1} \frac{x_a^2 + y_a^2 - 2}{2} \\ \psi_{i-1} &= \text{atan2}(y_a, x_a) + \cos^{-1} \frac{\sqrt{x_a^2 + y_a^2}}{2} \\ \psi_{i+1} &= -\psi_{i-2} - \psi_{i-1} - \psi_i \end{aligned} \right\} \quad (10)$$

where t is time, $t' = 6\tilde{t}^5 - 15\tilde{t}^4 + 10\tilde{t}^3$ is the fifth-dimensional curve given as a cam curve, $\tilde{t} = \frac{t-t_{st}}{t_{\psi}}$, and $t_{st} \leq t \leq t_{st} + t_{\psi}$. The robot can maintain a constant value of d_c , which is shown in Fig. 5(b), using (10). The shape of the connecting part can be determined freely as long as the relative distance d_c between the a th and b th wheels shown in Fig. 5(a) remains constant.

B. Start Condition for the Shift

It is assumed here that the robot does not move backwards when climbing up and down the stairs. The robot can determine whether or not each wheel axle is contacting the tread surface using the mounted sensor. The starting conditions for the shift process are designed as follows.

- 1) The b th module is not a connecting part.
- 2) The a th and b th wheels are located on the tread surface.

Condition 1 guarantees that the b th module is ready to be shifted. Condition 2 is necessary to maintain the static stability of the connecting part before and after the shifting process. If condition 2 is satisfied, then the connecting part is supported by the a th and b th wheels both during and after shifting. If the module that is included in the connecting part collides with the stairs before the a th wheel touches the tread surface, conditions 1 and 2 cannot be used. The available solutions in that case involve either improving the shape of the connecting part or using the $(a-1)$ th wheel rather than the a th wheel for condition 2 in Fig. 5(a).

C. Geometric Conditions of Stairs

The geometrical conditions that the stairs must satisfy when the robot climbs the stairs based on the starting conditions described in the previous subsection are described in the following. As shown in Fig. 6, we let Δ_a be the distance between the contact point and the rear end module of the connecting part in the x -direction as the robot moves forward until the robot and stair make contact. To satisfy condition 2, the $(j-1)$ th tread must satisfy the condition

$$d_{j-1} \geq L + \Delta_a \quad (11)$$

where Δ_a depends not only on the riser height but also on the module length L and the link geometry of the robot.

D. Velocity Constraint

If the robot moves forward appreciably during the shift, there may be a collision between the robot and stairs. It is thus desirable that the forward motion of the robot be restricted as much as possible during the shift process. However, there are cases where the wheel is determined to exist on the tread surface before it actually touches the tread surface as a result of the inaccuracy and noise error of the sensor used. In addition, because of the wheel slippage, the wheels [i.e., the a th and b th wheels in Fig. 5(a)] that should support the connecting part during the shift may move backwards and may then fall off the tread in some cases. In such cases, if the robot does not move forward, there is then the possibility that the balance will be lost during the shift. To solve these problems, the robot moves forward by a specific distance during each shift. Let Δ_L be the distance by which the robot moves forward during the shift. The forward velocity v during the shift can then be designed as

$$v = \Delta_L / t_{\psi}. \quad (12)$$

E. Climbing Down the Stairs

Fig. 5(c) depicts the shift in the connecting part when climbing down the stairs. The motion is similar to a reverse playback of the process of climbing up the stairs. The specific equation for the joint angle is omitted, and differences from the case of climbing up the stairs are described. The start conditions for climbing down the stairs are designed as follows.

- 1) The b' th module is not the connecting part.
- 2) The a' th and b' th wheels are located on the tread surface.
- 3) The c th wheel is not contacting the tread surface.

Conditions 1 and 2 here correspond to conditions 1 and 2 of Section III-B, respectively. Condition 3 represents the trigger for the start of the shift process. In climbing down the stairs, if the robot shifts with $\Delta_L = 0$ in (12), then the connecting part may contact the tread as shown in (B) in Fig. 5(d). In this case, the robot must move forward during the shift using an appropriate value of Δ_L , as shown in (B') in Fig. 5(d). The appropriate Δ_L depends on the wheel diameter, the riser height, and the link geometry of the robot.

If the b' th and c th wheels are changed to the $(b'+1)$ th and $(c+1)$ th wheels, respectively, the robot can then shift the connecting part without collision with $\Delta_L = 0$, but the required tread depth for the shift is increased. To match the required tread depths of the stair when climbing up and down, it is necessary to use the grounding and ungrounding of the c th wheel, as shown in Fig. 5(c), as the trigger for the shift when climbing down the stairs.

IV. DEVELOPMENT OF AN ARTICULATED MOBILE ROBOT

Fig. 7 depicts the developed robot, which is called *T² Snake-3* and has the parameters given in Table I. The robot is battery powered and can be controlled remotely on a wireless basis.

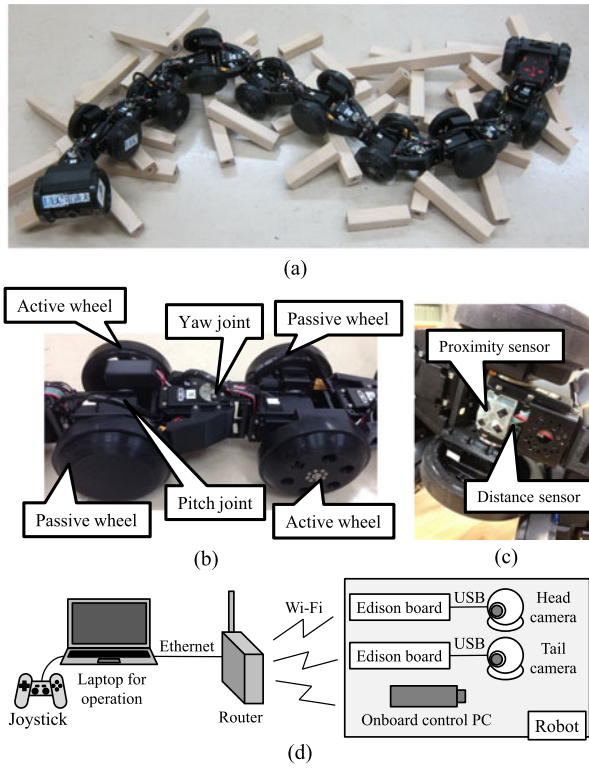


Fig. 7. Articulated mobile robot T^2 Snake-3. (a) Overall view, (b) enlarged view, (c) bottom view, and (d) complete system.

TABLE I
PARAMETERS OF THE ARTICULATED MOBILE ROBOT

Number of modules n	9
Link length l [mm]	90.5
Module length L [mm]	181
Total size ($H \times W \times L$) [mm]	$120 \times 150 \times 1729$
Wheel radius r [mm]	50
Joint angle range (Pitch) [deg.]	$ \psi_i \leq 113$
Joint angle range (Yaw) [deg.]	$ \phi_i \leq 65$
Total mass [kg]	9.2
Battery life [min.]	about 80

A. Mechanical Design

The joint configuration of the robot is the same as that of the model shown in Section II. Dynamixel servos (ROBOTIS Inc.) are used as actuators for both the joints and active wheels. MX-106R and MX-28AR models are used for the joints and wheels, respectively. To enable stair climbing when using the proposed method, the robot must have a wide pitch joint angle range that exceeds $\pm 90^\circ$, as in [23]. The range of the pitch joint angle is therefore designed to be as wide as possible.

In the stair climbing case, using the proposed method, the robot must have active wheels to be propelled because it does not undulate laterally in the manner of a snake robot, such as that of [24]. Additionally, the contact and noncontact conditions between the wheels and the tread surface change frequently. If at least one wheel on each axle is an active wheel, the robot can then generate propulsion because there are always wheels in contact with the tread. Therefore, one wheel on each axle is

the active wheel, while the other wheel is a passive wheel, as shown in Fig. 7(b). The wheels are active on both sides at the tail axle only. The mounting of these passive wheels is intended to reduce the overall size of the robot. The passive wheels are structured so that only the outer ring of each wheel rotates, and space for a battery is provided inside the passive wheel.

At the head and tail of the robot, cameras that obtain images, Intel Edison Modules that acquire these camera images and transmit them wirelessly, and batteries are installed. At the tail, an Intel Compute Stick is installed to act as the onboard control personal computer (PC) for the robot.

The robot has an infrared distance sensor and a proximity sensor [34] located at the bottom of its body to measure the distance and inclination angle between the robot and surrounding plane, as shown in Fig. 7(c). The robot can determine whether the wheel makes contact with the ground according to distance information with respect to the tread surface during stair climbing. Specifically, the contact between each pair of wheels and the stair tread surface is detected by comparing the distance information measured by the sensor with the threshold value.

B. Communication System Design

The communication system that connects the onboard control PC, actuators, and sensors comprises a master microcontroller unit (MCU) and multiple slave MCUs, and the actuators and sensors are connected to each of the slave MCUs. A SEED-MS1A (THK Co., Ltd.) is used as the MCU in this case. Communications are made between the master MCU and slave MCUs via a controller area network, while those between the master MCU and the control PC use an RS-485 system. The master MCU sends commands from the control PC to each slave MCU, collects data from the actuators and sensors that are connected to each slave MCU, and sends the resulting data to the control PC. Each slave MCU communicates with the corresponding actuator and reads the sensor data according to commands from the master MCU, and transmits the acquired data to the master MCU. During communications with the actuator, the slave MCU transmits the target angle and target speed, and receives the joint angle, current, and error information in return.

C. Complete System

Fig. 7(d) shows the complete system. The laptop used for system operation, the onboard control PC, and the Edison boards are all connected to the same network, which is managed by a router. The operator sends commands to the robot using a joystick. The operating system of the laptop for system operation is Ubuntu 14.04, and it acquires the joystick information via the robot operating system (ROS) and transmits this information to the control PC. The control PC then controls the robot by calculating the control inputs based on the operator's commands. The control PC uses the Windows 10 operating system, and the control PC uses MATLAB 2015b with the Robotics System Toolbox to control the robot and to communicate with the ROS network. Robot information (e.g., joint angle) acquired by the control PC is then transmitted to the operator's PC via the network.

TABLE II
BASIC MOBILITY PROPERTIES OF THE PROPOSED ROBOT

Propulsion speed [mm/s]	250
Maximum trench width [mm]	550
Maximum step height without riser [mm]	675
Maximum step height with riser [mm]	1030
Minimum width of L-shaped path [mm]	250

V. EXPERIMENT

A. Basic Performance Using FTL Control

The basic performance characteristics of the proposed robot are given in Table II. During the performance evaluation experiment, the method described in Section II was used to steer the robot, and the operator provided steering commands while looking at the robot itself rather than a camera image. The status of the experiments is illustrated in Fig. 8. While there are incomplete numbers in Table II because of the limitations of the experimental environment, the maximum sizes of both a trench and a step and the minimum size of an L-shaped path in Table II are the highest performance values that were obtained from the experiments, which were performed with increments of 50 mm. In the case of climbing a step with a riser, the wheels of the robot that were in contact with the riser surface not only generated propulsion but also reduced the load on the pitch joint, and it was possible to suppress tendencies to fall backward. As a result, the robot was able to climb to a higher step when compared with the case of climbing a step without a riser surface. This performance means that the robot can climb onto a typical desk if there is a riser surface available. Table III compares the step climbing performances of snake-like articulated mobile robots. The table shows that the proposed robot has a high step climbing performance. Even when the total length of the robot is considered, the performance of the proposed robot exceeds that of the rescue robot Soryu V [7], which has three crawler units that are serially connected by joints.

As a demonstration of the terrain-following performance achieved by manual intermittent adaptation, an operation to climb over a tool box with a height of 0.24 m was performed as shown in Fig. 8(b). The operator observed the robot motion visually and provided a command to steer the head and zero the torque. The joints at which the torque was zeroed are all pitch joints. The operator then applied a manual adaptation using the proposed method between $t = 17$ s and $t = 22$ s. At $t = 17$ s, many of the parts of the robot were lifted off the ground, and a heavy load was thus applied to the joints. In contrast, the body posture at $t = 22$ s was improved by the terrain adaptation procedure. From Fig. 8(c), the summation of the absolute value of the current of the joint decreased greatly to almost zero at the time of manual adaptation. After the robot adapted to the terrain, FTL operation resumed and the robot was able to climb over the box. Moreover, the robot was able to pass through a field of crossing ramps [35] using the proposed terrain-following method as shown in Fig. 8(d). In the field, there were cases that the joint was stuck after terrain following, but the robot was able

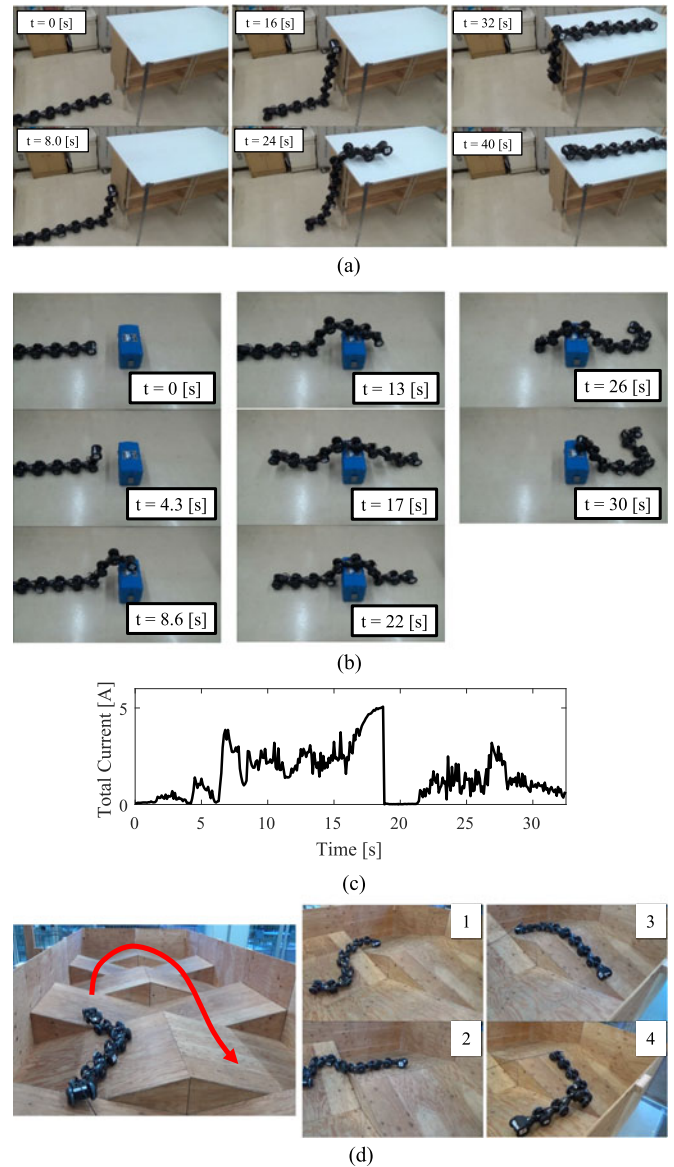


Fig. 8. Photographs of the experiments. (a) Climbing a step with a riser (height: 1030 mm). (b) Climbing a box using the proposed method. (c) Summation of the absolute value of the current of the joint when climbing a box. (d) Crossing ramps [35].

to recover from the stuck state by performing the traveling wave motion of the pitch angles described in Section II-F.

Note that the operation is performed manually, and the results thus depend on the skill of the operator. Establishment of an autonomous or semi-autonomous control method that is independent of the skill of the operator will be a subject of future work.

B. Semi-Autonomous Stair Climbing

The operator is only required to adjust the height of the foremost connecting part because the robot shifts this part from its head to its tail. The operator then adjusts the height while confirming the inclination outputs of both the proximity sensor and the distance sensor that are attached to the bottom of

TABLE III
COMPARISON OF STEP CLIMBING PERFORMANCES OF ARTICULATED MOBILE ROBOTS

Feature\Robot	T ² Snake-3	Soryu V[7]	OT-4[9]	RT04-NAGA[15]	ACM-R4.2[12]	ACM-R8[14]
Total length [mm]	1729	1160	940	1255	983	2010
Maximum step height with riser [mm]	1030	640	400	500	370	600
[%] of robot length	60	55	43	40	38	30
Maximum degree stairway [deg.]	54.5	—	40	40	—	30
(Rise/Tread) [mm]	(350/250)	—	—	(200/240)	—	(150/260)

the robot's body. A constant distance between the center of the dummy wheel and the tread surface is maintained by pressing the dummy wheel against the tread surface. Thus, a distance between the center of the head and the tread surface is maintained constant.

Fig. 9(a) and (b) shows how the robot climbed up and down stairs that were composed of risers and treads with various heights and depths. The operator manually adjusted the height of the robot's head while looking at the sensor value, and then, moved the robot forward. The robot started the shift in the connecting part at an appropriate time based on the available sensor information. As a result, the robot was able to ascend and descend stairs without becoming stuck.

If condition (11) is satisfied, the robot will also be able to climb steep stairs using the proposed method. Fig. 9(c) depicts the case of climbing steep stairs that were sloped at 54.5° , and which the other articulated mobile robots cannot climb, as shown in Table III. If the number of modules that make up the connecting part is increased, the robot can then climb even steeper stairs. However, it is noted that there are limits to the number of modules that can be used for the connecting part that depend on the actuator torque. Additionally, the parameters of the steep stairs used in this experiment do not conform to the ISO 14122-3 standard [36]. However, because the conditions for stair climbing when using the proposed method have been clarified in the aforementioned section, if the module length L of the robot can be reduced, then the robot will be able to climb steep stairs that satisfy the ISO standard.

Fig. 9(d) shows an experiment that was conducted on stairs at the University of Electro-Communications, Tokyo, Japan. On these stairs, the robot changed its target angle using the method described in Appendix A because the target angle exceeded the limit value given in Table I. The riser height is the same for all stairs and the operator only had to adjust the riser height for the first stair. The robot was able to climb up ten steps of these stairs in 205 s because the height adjustment was only required once and the adjustment time was short.

We then considered the case that the robot moved on stairs in an oblique direction that was not parallel to x in the xy plane. If the stair width is low, there is a risk that the robot will fall toward the side of the stairs or will collide with a wall at the side of the stairs because the robot cannot be steered using the proposed stair climbing method. It is thus desirable that the robot enters the stairs in parallel with the x -axis direction as much as possible. For example, if the posture of the robot before it climbs a stair is changed appropriately using the method of [37], the risks of falling and collision with a wall can be reduced.

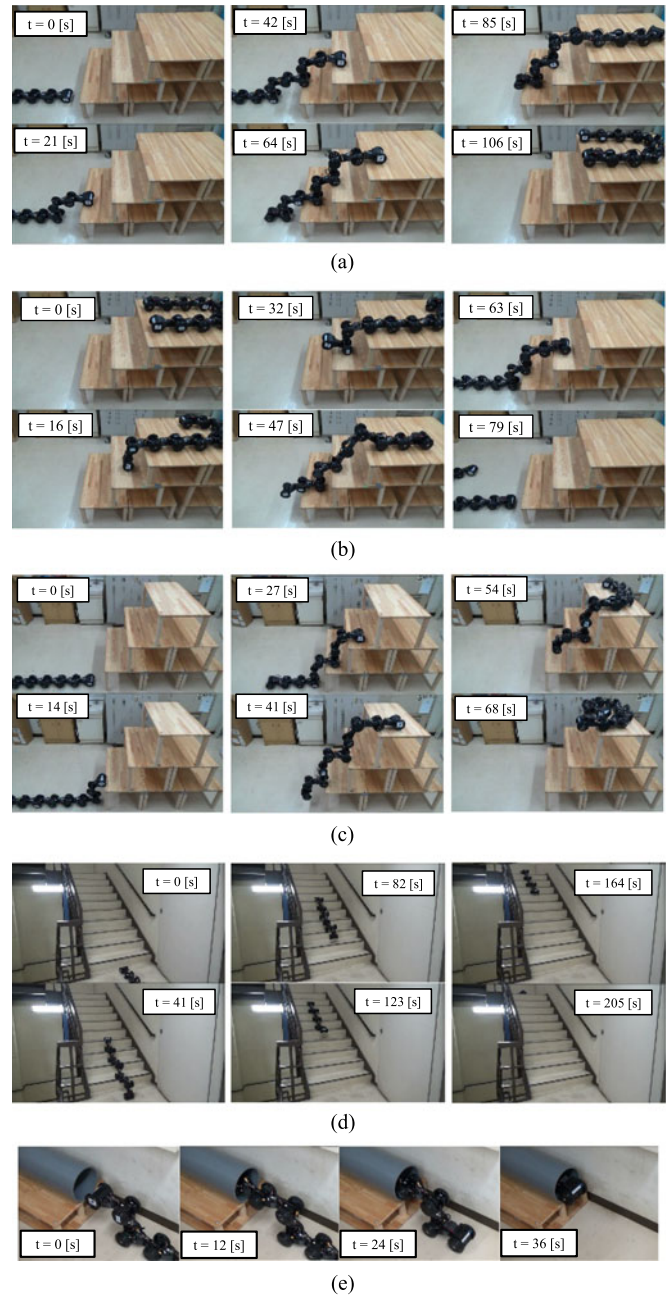


Fig. 9. Photographs of the robot using the semi-autonomous stair climbing method. (a) Climbing up stairs, where $h_1 = 0.25$, $h_2 = 0.25$, $h_3 = 0.22$, $d_1 = 0.3$, and $d_2 = 0.27$. (b) Climbing down stairs, where $h_1 = 0.25$, $h_2 = 0.25$, $h_3 = 0.22$, $d_1 = 0.3$, and $d_2 = 0.27$. (c) Climbing up steep stairs, where $d_j = 0.25$, $h_j = 0.35$, and the rise/tread combination is 54.5° . (d) Climbing up actual stairs, where $d_j = 0.255$ and $h_j = 0.185$. (e) Entering a pipe (pipe inner diameter: 194 mm).

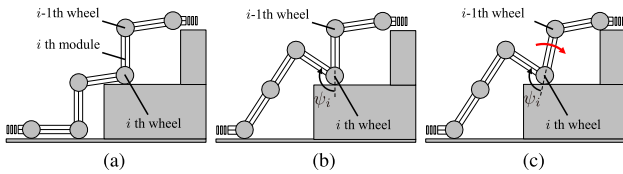


Fig. 10. Joint limit avoidance. (a) Before the shift, (b) during shifting, and (c) during shifting while avoiding the joint limit.

If the robot can detect the height of the stair with a laser range sensor mounted at the head of the robot as in [21], it seems that the robot will be able to automatically ascend and descend stairs without an operator, which is a topic of future work.

The stair climbing method can also be applied to allow the robot to enter a horizontal pipe, as shown in Fig. 9(e). The distance between the ground plane and the inner wall of the pipe is 180 mm. In the case that the robot enters the pipe using the FTL method, it is possible that the joint part of the robot will collide with the pipe because the following part of the robot may pass along a path that differs greatly from that followed by the robot's head owing to slippage of the wheel. In contrast, in the case that the stair climbing method is used, even if there is slippage of the wheel, the robot can still enter the pipe because the robot can shift the connecting part at an appropriate time according to sensor information.

VI. CONCLUSION

With the aim of realizing an articulated mobile robot that is capable of not only movement for search and rescue at disaster sites but also climbing steep stairs for inspecting industrial plants, this paper presented a basic 3-D steering method with terrain following based on intermittent manual adaptation and a semi-autonomous stair climbing method that uses sensor information. An articulated mobile robot was then developed, and the effectiveness of the proposed control method was demonstrated experimentally. The developed robot can move along a narrow path using its elongated body, adapt to the surrounding terrain, climb a 1-m-high step, and climb steep stairs. On the basis of these characteristics, the robot will be able to enter and overcome many different terrains, and it is likely to be useful for the inspection of houses and industrial plants.

Future work will involve the proposal of a control method that assumes use on terrains such as a spiral staircase, the interior and exterior of a pipe, and rubble, with the aim of expanding the range of terrain through which the robot can move.

APPENDIX A JOINT LIMIT AVOIDANCE

When the rotation angle required for shifting the connecting part exceeds the robot's limit angle, it is necessary to modify the shift motion. Fig. 10 depicts an example of such modification. We consider the case that the i th pitch joint reaches the limit angle. Let ψ_L be the limit angle and let ψ_{id} be the target angle of the i th pitch angle. In the case that $\psi_{id} > \psi_L$, we design the i th pitch angle to be $\psi_i = \psi_L$ and tilt the i th module forward by $\alpha = \psi_{id} - \psi_L$. The values of ψ_{i-1} and ψ_{i-2} are then modified so

that the $(i-2)$ th wheel continues to touch the tread surface. As a result, the robot can shift the connecting part while avoiding the joint limit. The part of the robot from the head to the i th module temporarily moves forward while avoiding this joint limit. Note that success in stair climbing is not guaranteed because there is always the possibility that the part will come into contact with the stairs when it is temporarily moved forward.

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Motoyasu Tanaka (S'05–M'12) received the B.Eng., M.Eng., and Ph.D. degrees in engineering from the Department of Mechanical Engineering and Intelligent Systems, University of Electro-Communications, Chofu, Japan, in 2005, 2007, and 2009, respectively.

From 2009 to 2012, he worked at Canon, Inc., Tokyo, Japan. He is currently an Associate Professor with the Department of Mechanical and Intelligent Systems Engineering, University of Electro-Communications. His research inter-

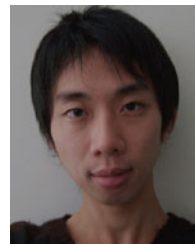
ests include biologically inspired robotics and dynamics-based nonlinear control.

Dr. Tanaka was the recipient of the IEEE Robotics and Automation Society Japan Chapter Young Award from the IEEE Robotics and Automation Society Japan Chapter in 2006, and the Best Poster Award at SWARM2015: The First International Symposium on Swarm Behavior and Bio-Inspired Robotics in 2015.



Mizuki Nakajima received the B.Eng. and M.Eng. degrees in engineering from the Department of Mechanical Engineering and Intelligent Systems, University of Electro-Communications, Tokyo, Japan, in 2014 and 2016, respectively, where he is currently working toward the Ph.D. degree.

His research interests include the development and control of snake robots.



Yosuke Suzuki (M'17) received the B.Eng., M.Eng., and Ph.D. degrees in engineering from the Tokyo Institute of Technology, Tokyo, Japan, in 2005, 2007, and 2010, respectively.

He is currently an Assistant Professor with the Faculty of Mechanical Engineering, Institute of Science and Engineering, Kanazawa University, Kanazawa, Japan. His research interests include modular robots, tactile and proximity sensors, and robotic grasping.



Kazuo Tanaka (S'87–M'91–SM'09–F'14) received the B.S. and M.S. degrees in electrical engineering from Hosei University, Tokyo, Japan, in 1985 and 1987, respectively, and the Ph.D. degree in systems science from the Tokyo Institute of Technology, Tokyo, in 1990.

He is currently a Professor with the Department of Mechanical and Intelligent Systems Engineering, University of Electro-Communications, Tokyo. He was a Visiting Computer Scientist at the University of North Carolina

at Chapel Hill, Chapel Hill, NC, USA, in 1992 and 1993. He is the author of two books and co-author of 17 books. Recently, he co-authored (with Hua O. Wang) *Fuzzy Control Systems Design and Analysis: A Linear Matrix Inequality Approach* (Wiley-Interscience, 2001). His research interests include intelligent systems and control, nonlinear systems control, robotics, brain-machine interfaces, and their applications.

Dr. Tanaka was the recipient of the Best Young Researcher Award from the Japan Society for Fuzzy Theory and Systems in 1990; the Outstanding Paper Award at the 1990 Annual North American Fuzzy Information Processing Society (NAFIPS) Meeting in Toronto, ON, Canada, in 1990; the Outstanding Paper Award at the Joint Hungarian-Japanese Symposium on Fuzzy Systems and Applications in Budapest, Hungary, in 1991; and the Best Young Researcher Award from the Japan Society for Mechanical Engineering in 1994. He was also the recipient of the Outstanding Book Award from the Japan Society for Fuzzy Theory and Systems in 1995, the 1999 IFAC World Congress Best Poster Paper Prize in 1999, the 2000 IEEE Transactions on Fuzzy Systems Outstanding Paper Award in 2000, and the Best Paper Selection at 2005 American Control Conference in Portland, OR, USA, in 2005. He is currently serving as an Associate Editor for *Automatica* and the IEEE TRANSACTIONS ON FUZZY SYSTEMS, and is on the IEEE Control Systems Society Conference Editorial Board.